Huawei Access Node JAY-ACC-60 Routing Protocol Flap SOP

Node: JAY-ACC-60 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5325

Description

This Standard Operating Procedure (SOP) addresses a major failure scenario on Huawei Access Node JAY-ACC-60, specifically within the Routing Protocol Flap failure domain. The failure domain encompasses BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, all of which are critical components of the node's routing protocol functionality. The protocol flap phenomenon can be attributed to the dynamic nature of routing protocols, where periodic changes in the network topology or configuration can cause the node to rapidly switch between different routing states. This can lead to an excessive number of routing protocol events, resulting in a significant increase in CPU utilization and potentially causing instability within the network. On the Huawei access node, the routing protocol flap is typically triggered by an OSPF adjacency loss or BGP route churn, which in turn can be caused by various factors such as link flapping, routing policy changes, or software bugs. The root cause analysis of this failure domain is complex and requires a rigorous understanding of the network architecture, routing protocols, and Huawei-specific configuration and troubleshooting procedures. This SOP aims to provide a comprehensive framework for diagnosing and resolving the Routing Protocol Flap failure scenario on Huawei Access Node JAY-ACC-60.

Pre-requisite Checks

- Ensure that the Jayanagar Metro Site is accessible and that the necessary remote hands are dispatched to the site for physical intervention.

- Verify the status of the redundant link to ensure that it is stable and not contributing to the failure scenario.

- Access the Huawei access node's command-line interface (CLI) as a privileged user.

Potential Root Causes

Cause	Probability
OSPF Adjacency Loss due to Link Flapping	High
BGP Route Churn due to Routing Policy Changes	Medium
ISIS LSP Flooding due to Software Bugs	Low

Diagnostic Commands & Expected Outputs

```
Command: display ospf interface, Expected Output: Interface status: Up, Adjacency status: Down, Description: This command isolates the OSPF adjacency loss issue and verifies the interface status and adjacency state.
```

```
Command: display bgp routing-table, Expected Output: Route churn detected, Number of routes: 1000, Description: This command isolates the BGP route churn issue and verifies the number of routes affected.

Command: display isis lsp, Expected Output: LSP flooding detected, Number of LSPs: 500, Description: This command isolates the ISIS LSP flooding issue and verifies the number of LSPs affected.
```

Resolution Steps

Step 1: Step: Stabilize the OSPF Adjacency, Command: undo ospf interface, Description: This command resolves the OSPF adjacency loss issue by removing the OSPF interface configuration.

Step 2: Step: Re-Configure the BGP Routing Policy, Command: configure terminal bgp routing-policy, Description: This command resolves the BGP route churn issue by re-configuring the BGP routing policy.

Step 3: Step: Clear ISIS LSP Flooding, Command: clear isis lsp, Description: This command resolves the ISIS LSP flooding issue by clearing the LSP flooding.

Post-Check Verification

```
Command: display ospf interface, Expected Output: Interface status: Up, Adjacency status: Up, Description: This command verifies that the OSPF adjacency is stable.

Command: display bgp routing-table, Expected Output: Route churn detected, Number of routes: 0, Description: This command verifies that the BGP route churn issue is resolved.

Command: display isis lsp, Expected Output: LSP flooding detected, Number of LSPs: 0, Description: This command verifies that the ISIS LSP flooding issue is resolved.
```

Roll-back Procedures

- Step: Restore the Original OSPF Configuration, Command: configure terminal ospf interface, Description: This command restores the original OSPF configuration if the resolution step fails.

- Step: Restore the Original BGP Routing Policy, Command: configure terminal bgp routing-policy, Description: This command restores the original BGP routing policy if the resolution step fails.

- Step: Restore the Original ISIS Configuration, Command: configure terminal isis, Description: This command restores the original ISIS configuration if the resolution step fails.